

Axel VandenHeuvel

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EDUCATION

University of Colorado Boulder

Boulder, CO

Bachelor of Science in Computer Science

May 2026

- GPA: 3.66/4.0 — Computer Science GPA: 3.96/4.0

EXPERIENCE

Onboard Vision-Language Models for Deep Space Spacecraft

August 2025 – July 2026

NASA JPL Sponsored Capstone | C++, VLMs, Linux | [GitHub Link](#)

Boulder, CO

- Authored PiVision, a C++ library and CLI on llama.cpp's multimodal API for running quantized VLMs on CPU-only ARM64 boards; its timing instrumentation produced the runtime data for the team's benchmark of 13 Gemma 3 and Ministral 3 variants on a Raspberry Pi 5.
- Co-authored a peer-reviewed paper accepted to IEEE SMC-IT/SCC 2026 with a JPL mentor, evaluating output quality with an LLM-as-judge rubric and mapping model choice to mission latency budgets.

Cloud Solutions / Azure Intern

June 2025 – August 2025

Tenet Healthcare

Dallas, TX

- Reclaimed unused Microsoft 365 and Azure licensing through automated audit and analysis pipelines, contributing to more than \$500K in projected annual cost savings.
- Integrated the DocuSign API to automate compliance reminder workflows for outstanding documents.
- Built PowerShell automation to detect misconfigurations and manage entitlement workflows across 10,000+ users.

PUBLICATIONS

K. Suoo, J. Li, P. Hunt, **A. VandenHeuvel**, N. Khazam, P. Barger, P. Secrest, and M. Llopis. "Evaluating Quantized Vision-Language Models for Spacecraft-Relevant Inference on Constrained Hardware." *IEEE Space Mission Challenges for Information Technology / Space Computing Conference (SMC-IT/SCC)*, 2026. Accepted.

PROJECTS

Local Dictation App for macOS | *Swift, CoreML, Apple Foundation Models* | [GitHub Link](#)

August 2026

- Built and shipped a macOS speech-to-text app via Homebrew with fully local AI — no cloud, no API keys.
- Runs NVIDIA's Parakeet v3 speech model on the Apple Neural Engine at ~50x real time, then refines the transcript with Apple's on-device foundation model, deterministic cleanup rules, and fuzzy user-vocabulary correction.

Lift-Tracker iOS App | *Swift* | [App Store Link](#) | [GitHub Link](#)

March 2026

- Shipped a native weightlifting tracker to the iOS App Store: calendar-based logging, reusable workout templates, a 154-movement library, and per-movement progress charts with automatic weight and rep PR detection.

GraphRAG Knowledge System | *Python, FastAPI, TypeScript* | [GitHub Link](#)

March 2026

- Co-built (team of 4) a hybrid vector/graph RAG desktop app that ingests notes (Markdown, PDF, Notion), extracts and canonicalizes concepts with an LLM, and renders them as an interactive 3D force-directed knowledge graph inside a brain mesh.
- Stabilized the FastAPI backend (shared Kuzu engine, per-request connections, async endpoints) and built the ingestion pipeline: PDF/zip upload, duplicate detection, Notion importer, and a Milkdown WYSIWYG editor with tabs.

TECHNICAL SKILLS & INTERESTS

Languages: Python, TypeScript / JavaScript, C++, Swift

AI Systems: RAG & hybrid retrieval, Ollama / llama.cpp, on-device inference, model evaluation & benchmarking

Cloud & Enterprise: Azure, Microsoft 365 administration, PowerShell, DocuSign API, REST integrations

Frameworks & Libraries: Flask, NumPy, pandas, SwiftUI, FastAPI

Tools: Git, Claude Code, Raspberry Pi, Codex, Neovim, tmux

Interests: Hiking, Camping, Weightlifting, Mechanical Keyboards, Astronomy